

ALBERT EINSTEIN

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RESEARCH INTERESTS

Theoretical physics, general relativity, quantum mechanics, statistical mechanics, unified field theory

EDUCATION

ETH Zurich (Swiss Federal Polytechnic) Ph.D. in Physics	Zurich, Switzerland 1900–1905
ETH Zurich (Swiss Federal Polytechnic) Diploma in Mathematics and Physics	Zurich, Switzerland 1896–1900
Aargau Cantonal School Secondary Education Certificate	Aarau, Switzerland 1895–1896

EXPERIENCE

Institute for Advanced Study Professor of Theoretical Physics	Princeton, NJ, USA 1933–1955
• Conducted research on unified field theory and cosmology • Mentored doctoral and post-doctoral researchers • Published 33 papers during tenure	
University of Berlin Professor of Theoretical Physics	Berlin, Germany 1914–1933
• Developed foundations of general relativity theory • Elected to Prussian Academy of Sciences • Led Kaiser Wilhelm Institute for Physics	
Swiss Patent Office Technical Expert, Second Class	Bern, Switzerland 1902–1909
• Published "Annus Mirabilis" papers (1905) • Developed special relativity and mass-energy equivalence • Contributed to quantum theory foundations	

HONORS & AWARDS

• Nobel Prize in Physics, <i>Royal Swedish Academy of Sciences</i>	1921
• Copley Medal, <i>Royal Society of London</i>	1925
• Max Planck Medal, <i>German Physical Society</i>	1929
• Franklin Medal, <i>Franklin Institute</i>	1935
• Foreign Member, <i>Royal Society of London</i>	1921
• Member, <i>National Academy of Sciences</i>	1922

PUBLICATIONS

* Equal contribution † Corresponding author

- [1] **Albert Einstein**^{*}, Boris Podolsky^{*†}, and Nathan Rosen[†]. “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?” In: *Physical Review* 47.10 (1935). EPR paradox paper, pp. 777–780.
- [2] **Albert Einstein**. “Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie”. In: *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften* (1917). Introduction of cosmological constant, pp. 142–152.
- [3] **Albert Einstein**. “Die Grundlage der allgemeinen Relativitätstheorie”. In: *Annalen der Physik* 49.7 (1916). General theory of relativity, pp. 769–822.
- [4] **Albert Einstein**. “Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig?” In: *Annalen der Physik* 18.13 (1905). Mass-energy equivalence, $E = mc^2$, pp. 639–641.
- [5] **Albert Einstein**. “Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt”. In: *Annalen der Physik* 17.6 (1905). Nobel Prize paper on photoelectric effect, pp. 132–148.
- [6] **Albert Einstein**. “Zur Elektrodynamik bewegter Körper”. In: *Annalen der Physik* 17.10 (1905). Special theory of relativity, pp. 891–921.

TECHNICAL SKILLS

Theoretical Physics: Relativity Theory, Quantum Mechanics, Statistical Mechanics, Thermodynamics
Mathematics: Differential Geometry, Non-Euclidean Geometry, Tensor Analysis, Differential Equations
Programming Languages: Fortran, Assembly (theoretical calculations)
Mathematical Tools: Analytical mechanics, Variational calculus, Complex analysis
Languages: German (native), English (fluent), French (working proficiency), Italian (basic)

SOFTWARE

- [relativity-simulator](#), *Personal project* 1920–1930
- [quantum-probability](#), *Collaborative research* 1924–1935

TALKS & PRESENTATIONS

- The Evolution of Physics, *University of Oxford* 1933
- The Unified Field Theory, *California Institute of Technology* 1931
- The Theory of Relativity, *Columbia University* 1921
- Space, Time and Gravitation, *King’s College London* 1921
- The Meaning of Relativity, *Princeton University* 1921

TEACHING

COURSE INSTRUCTOR

- Advanced Seminar in Theoretical Physics, *Institute for Advanced Study* 1933–1955
- Graduate Lectures in Relativity Theory, *University of Berlin* 1914–1933

GUEST LECTURES

- Lectures on Molecular Physics, *University of Zurich* 1909–1911

MENTORING

- Nathan Rosen, *PhD advisor* 1932–1935
- Peter Bergmann, *PhD advisor* 1933–1936
- Bruria Kaufman, *PhD co-advisor* 1948–1951
- Leopold Infeld, *Postdoc* 1936–1938
- Valentine Bargmann, *PhD committee* 1936–1938
- John Wheeler, *PhD committee* 1933–1935
- Abraham Pais, *Visiting researcher* 1946–1947

SERVICE

CONFERENCE ORGANIZATION

- Solvay Conference on Physics, *Organizing Committee* 1927, 1930
- International Congress of Mathematicians, *Advisory Board* 1928
- Prussian Academy Physics Colloquium, *Organizing Committee* 1920–1933

PEER REVIEW

- Annalen der Physik: 1905–1933
- Physical Review: 1935–1955
- Proceedings of the Royal Society: 1920–1950
- Zeitschrift für Physik: 1920–1933

DEPARTMENTAL & UNIVERSITY SERVICE

- Board of Governors, *Hebrew University of Jerusalem* 1925–1955
- Faculty Senate, *Institute for Advanced Study* 1933–1955
- Academic Committee, *Kaiser Wilhelm Institute* 1914–1933
- Admissions Committee, *ETH Zurich* 1912–1914

COMMUNITY SERVICE

- Emergency Committee of Atomic Scientists, *Chairman* 1946–1955
- International Rescue Committee, *Advisory Board* 1933–1955
- League of Nations Committee on Intellectual Cooperation, *Member* 1922–1932
- World Peace Congress, *Keynote Speaker* 1949